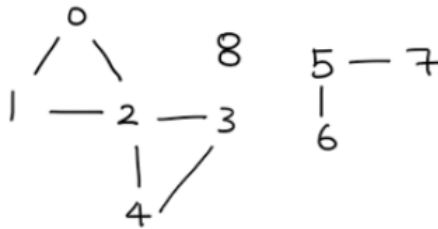


L2: Knowledge Check

Multiple choice 1 point

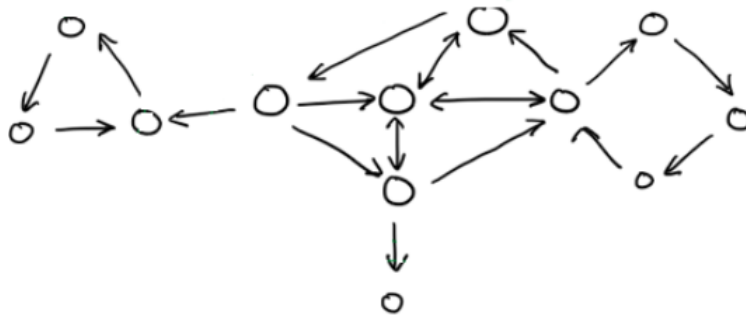
Which of the following statements about this graph is **False**?



- ☐ Sum of all node degrees = 8
- ☐ Number of edges = 8
- ☐ Graph density = $2/9$
- ☐ Max number of possible edges = 36

Multiple choice 1 point

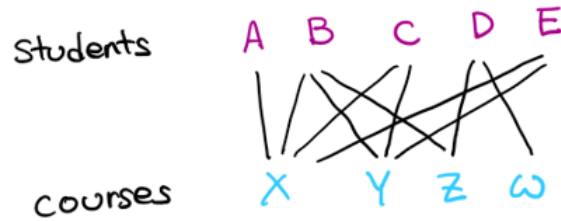
How many Strongly Connected Components are there in the following graph? (note that an individual node is, by convention, a strongly connected component)



- 3
○ 4
○ 2
○ 1

3 Multiple choice 1 point

Which of the following statements about this below bipartite graph is **False**?



- ☐ There is a cycle of length 5
- ☐ The bibliographic coupling of B and D is 1
- ☐ There is a cycle of length 4
- ☐ The weight of the edge AB of the one-mode projection on students is 1

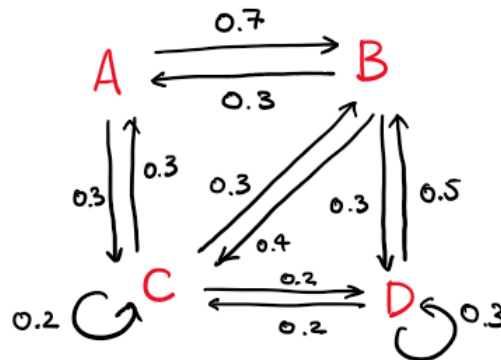
4 Multiple choice 1 point

Which of the following statements is **False**?

- ☐ Dijkstra's algorithm can be used for any weighted graph with positive or negative weights to calculate the shortest path from a node to any other node in the graph.
- ☐ BFS can be used for computing all connected components of an undirected graph.
- ☐ BFS can be used for any unweighted graph to calculate the shortest path from a node to any other node in the graph.
- ☐ Tarjan's algorithm can be used to compute the strongly connected components of a directed network.

5 Multiple choice 1 point

This plot shows a 4-node network with the associated transition matrix. The random walker is initially at node A. What is the stationary distribution for this random walk?



1. $[0.32 \ 0.181 \ 0.282 \ 0.218]^T$
2. $[0.181 \ 0.32 \ 0.282 \ 0.218]^T$
3. $[0.282 \ 0.218 \ 0.181 \ 0.32]^T$
4. $[0.181 \ 0.218 \ 0.282 \ 0.32]^T$

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

6

Fill in the Blank 1 point

Answer true or false for each of the following statements.

1. Dijkstra's shortest path algorithm ($\mathcal{O}(m + n \log n)$) has better running time complexity than the Bellman-Ford algorithm ($\mathcal{O}(m \times n)$) on both **dense** and **sparse** non-negative weighted networks.

2. Directed acyclic graphs have a unique topological ordering.

3. In a social network, checking whether two people are friends requires fewer memory accesses if we use an adjacency matrix instead of an adjacency list.

4. Our in and out node strength definition for weighted directed graphs can be also applied to signed graphs where weights can be negative.

☐ False

☐ True

☐ True

☐ False

☐ False

☐ True

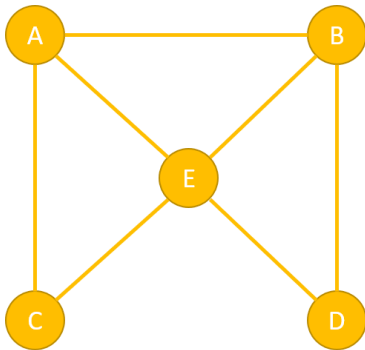
☐ True

☐ False

7

Multiple choice 1 point

How many walks of length 3 are there between node A and C?


☐ 3

☐ 4

☐ 5

☐ 6